

[REVIEW]: gimap: An R Package for Genetic Interaction Mapping in Dual-Target CRISPR Screens #9891

Open



editorialbot opened on Jan 23 · edited by editorialbot

Edits Collaborator

Submitting author: [@cansavvy](#) (Candace Savonen)
Repository: <https://github.com/FredHutch/gimap>
Branch with paper.md (empty if default branch):
Version: v1.1.1
Editor: [@fboehm](#)
Reviewers: [@kalyanidhusia](#), [@btmonier](#), [@tomsing1](#)
Archive: Pending

Status

JOSS Under Review

Status badge code:

```
HTML: <a href="https://joss.theoj.org/papers/33f30f34043821f41b527092a8f00a5f"></a>
Markdown: [! [status] (https://joss.theoj.org/papers/33f30f34043821f41b527092a8f00a5f/status.svg)]
(https://joss.theoj.org/papers/33f30f34043821f41b527092a8f00a5f)
```

Reviewers and authors:

Please avoid lengthy details of difficulties in the review thread. Instead, please create a new issue in the [target repository](#) and link to those issues (especially acceptance-blockers) by leaving comments in the review thread below. (For completists: if the target issue tracker is also on GitHub, linking the review thread in the issue or vice versa will create corresponding breadcrumb trails in the link target.)

Reviewer instructions & questions

[@kalyanidhusia](#) & [@btmonier](#), your review will be checklist based. Each of you will have a separate checklist that you should update when carrying out your review.

First of all you need to run this command in a separate comment to create the checklist:

```
@editorialbot generate my checklist
```

The reviewer guidelines are available here: https://joss.readthedocs.io/en/latest/reviewer_guidelines.html. Any questions/concerns please let [@fboehm](#) know.

✨ Please start on your review when you are able; the expectation is that it will take 2-4 weeks to complete. ✨

Checklists

[Checklist for @btmonier](#)

[Checklist for @tomsing1](#)



editorialbot assigned [fboehm](#) on Jan 23

editorialbot added [review](#) [TeX](#) [R](#) on Jan 23

 editorialbot mentioned this on Jan 23

📄 [PRE REVIEW]: gimap: An R Package for Genetic Interaction Mapping in Dual-Target CRISPR Screens #8797

 editorialbot added HTML Track: 2 (BCM) pre-2026-submission on Jan 23

editorialbot on Jan 23

Collaborator Author ...

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@editorialbot commands



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@editorialbot generate pdf



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Collaborator Author ...

Reference check summary (note 'MISSING' DOIs are suggestions that need verification):



✅ OK DOIs

- 10.1093/bioinformatics/btae469 is OK
- 10.1038/s41568-024-00763-x is OK
- 10.1371/journal.pgen.1008466 is OK
- 10.1016/j.celrep.2021.109597 is OK
- 10.1038/s41467-021-21478-9 is OK
- 10.1186/s13059-014-0554-4 is OK
- 10.1038/s41587-020-0437-z is OK

● SKIP DOIs

- No DOI given, and none found for title: DepMap Public 25Q2
- No DOI given, and none found for title: Babraham Bioinformatics - FastQC A Quality Control...

❌ MISSING DOIs

- 10.1111/j.2517-6161.1995.tb02031.x may be a valid DOI for title: Controlling the false discovery rate: a practical ...

❌ INVALID DOIs

- None



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Collaborator Author ...

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Collaborator Author ...

Repository Analysis Report

Code breakdown



Language	files	blank	comment	code
Text	6	2	0	393987
CSV	3	1	0	35058
HTML	67	4483	509	23995
R	20	363	1131	2179
Rmd	13	548	947	818
Markdown	10	217	7	488
CSS	3	98	52	442
JavaScript	3	64	32	256
YAML	6	34	15	166
TeX	1	11	0	126
XML	1	1	0	58
SVG	1	0	1	11
Dockerfile	1	3	3	9
SUM:	135	5825	2697	457593

Commit Count by Author



364 Candace Savonen
88 Kate Isaac
28 cansavvy
9 Howard Baek
5 Daniel J. Groso
5 sobrien29
2 ahberger
1 Davis Vaughan
1 Phoebe Parrish
1 marissafujimoto

Repository History

Repository age: First commit on January 03, 2024

Code distribution (top 3 48-hour windows):

Period	Insertions	% of Total	Signal	View Changes
Jul 29, 2024 - Jul 31, 2024	1849190	25.3%	●	View diff
Oct 22, 2024 - Oct 24, 2024	995090	13.6%	●	View diff
Jan 22, 2025 - Jan 24, 2025	602797	8.2%	●	View diff

GitHub Activity Metrics

Metric	Count
GitHub stars	1
Forks	2
GitHub contributors	9
Releases	1
Total issues	31
Total pull requests	75
Unique commenters/reviewers (excluding authors)	7



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Collaborator Author ...

Paper file info:



⚠ Wordcount for `paper.md` is **1803**

✓ The paper includes a `Statement of need` section



editorialbot on Jan 23

Collaborator

Author



License info:

● License found: `Creative Commons Zero v1.0 Universal` (Not [OSI approved](#))



fboehm on Jan 28



[@kalyanidhusia](#) [@btmonier](#) - please create your review checklists by following the instructions above. Please let me know if you encounter any issues. Thanks!



btmonier on Jan 28 · edited by btmonier

Edits



Review checklist for [@btmonier](#)

AI usage

- ✓ I confirm that I have read the [JOSS AI policy](#) and understand that:
 - AI may only be used for **non-substantive assistance** (e.g., grammar checking my review text, contextualizing public materials)
 - **All evaluative decisions**—including scoring, accept/reject recommendations, and assessments of originality, correctness, significance, and ethics compliance—**must be made exclusively by me**, not AI
 - I bear full responsibility for all judgments, recommendations, and technical claims in my review
 - I will disclose any AI assistance at the conclusion of my review

Conflict of interest

- ✓ I confirm that I have read the [JOSS conflict of interest \(COI\) policy](#) and that: I have no COIs with reviewing this work or that any perceived COIs have been waived by JOSS for the purpose of this review.

Code of Conduct

- ✓ I confirm that I read and will adhere to the [JOSS code of conduct](#).

General checks

- ✓ **Repository:** Is the source code for this software available at the <https://github.com/FredHutch/gimap?>
- ✓ **License:** Does the repository contain a plain-text LICENSE or COPYING file with the contents of an [OSI approved](#) software license?
- ✓ **Contribution and authorship:** Has the submitting author ([@cansavvy](#)) made major contributions to the software? Does the full list of paper authors seem appropriate and complete?
- ✓ **Substantial scholarly effort:** Does this submission meet the scope eligibility described in the [JOSS guidelines](#)
- ✓ **Data sharing:** If the paper contains original data, data are accessible to the reviewers. If the paper contains no original data, please check this item.
- ✓ **Reproducibility:** If the paper contains original results, results are entirely reproducible by reviewers. If the paper contains no original results, please check this item.
- ✓ **Human and animal research:** If the paper contains original data research on humans subjects or animals, does it comply with [JOSS's human participants research policy and/or animal research policy](#)? If the paper contains no such data, please check this item.

Functionality

- ✓ **Installation:** Does installation proceed as outlined in the documentation?
- ✓ **Functionality:** Have the functional claims of the software been confirmed?
- ✓ **Performance:** If there are any performance claims of the software, have they been confirmed? (If there are no claims, please check off this item.)

Documentation

- ✓ **A statement of need:** Do the authors clearly state what problems the software is designed to solve and who the target audience is?
- ✓ **Installation instructions:** Is there a clearly-stated list of dependencies? Ideally these should be handled with an automated package management solution.
- ✓ **Example usage:** Do the authors include examples of how to use the software (ideally to solve real-world analysis problems).
- ✓ **Functionality documentation:** Is the core functionality of the software documented to a satisfactory level (e.g., API method documentation)?
- ✓ **Automated tests:** Are there automated tests or manual steps described so that the functionality of the software can be verified?
- ✓ **Community guidelines:** Are there clear guidelines for third parties wishing to 1. Contribute to the software 2. Report issues or problems with the software 3. Seek support

Software paper

- ✓ **Summary:** Has a clear description of the high-level functionality and purpose of the software for a diverse, non-specialist audience been provided?
- ✓ **A statement of need:** Does the paper have a section titled 'Statement of need' that clearly states what problems the software is designed to solve, who the target audience is, and its relation to other work?
- ✓ **State of the field:** Do the authors describe how this software compares to other commonly-used packages?
- ✓ **Quality of writing:** Is the paper well written (i.e., it does not require editing for structure, language, or writing quality)?
- **References:** Is the list of references complete, and is everything cited appropriately that should be cited (e.g., papers, datasets, software)? Do references in the text use the proper [citation syntax](#)?



btmonier on Feb 3 · edited by btmonier

Edits ▾ ...

Hi [@cansavvy](#) - this looks like a useful package for the CRISPR community. The software is clear, easy to install and to use. Some minor comments I have before I can check the last boxes in my checklist:

Licensing Conflicts - In your repository, you state [CC0-1.0](#) (which is not [OSI approved](#)), but in your [DESCRIPTION](#) file, you state GPL-3.

Installation - In your README, you specify installation via the `remotes` package, but on the [provided website](#), you also mention installation through CRAN using `install.packages("gimap")`. I was unable to install it with the CRAN approach since it appears to be missing:

```
Warning message:
package 'gimap' is not available for this version of R

A version of this package for your version of R might be available elsewhere,
see the ideas at
https://cran.r-project.org/doc/manuals/r-patched/R-admin.html#Installing-packages
```



Community Guidelines - please set up issue tracking on this repo in case the community wants to reach out for possible bugs, feature requests, enhancements, etc.

Paper Submission - Some nitpicks:

- In [paper.bib](#) you specify an entry that is not referenced in the paper (i.e. the article tagged as `bioinformatics`)
- In the [paper.md](#) [front matter](#), you list one of the authors with the last name of "Groso", but in the [DESCRIPTION](#) file, you specify this individual's last name as "Grosso"
- The paper detailing the "Benjamini-Hochberg" procedure (tagged as `benjamini1995controlling` in your [paper.bib](#)) is missing a DOI (as flagged by the bot) - you should add some `doi` and `url` keys to this BibTeX entry, e.g.:

```
@article{benjamini1995controlling,
  title = {Controlling the false discovery rate: a practical and powerful approach to multiple testing},
  author = {Benjamini, Yoav and Hochberg, Yosef},
  journal = {Journal of the Royal Statistical Society: Series B (Methodological)},
  volume = {57},
  number = {1},
  pages = {289--300},
  year = {1995},
  publisher = {Wiley Online Library},
  doi = {10.1111/j.2517-6161.1995.tb02031.x},
  url = {https://doi.org/10.1111/j.2517-6161.1995.tb02031.x}
}
```



Example Usage - Looking at this code in the [quick start guide](#):

```
gimap_dataset <- gimap_dataset %>%  
  gimap_filter() %>%  
  gimap_annotate(cell_line = "HELA") %>%  
  gimap_normalize(  
    timepoints = "day"  
  ) %>%  
  calc_gi()
```

It seems that something has changed with the returned data and is causing downstream exceptions which needs to be addressed:

```
DepMap metadata format has changed - expected column 'stripped_cell_line_name' not found. Returning dataset without cell line annotation.  
Error in gimap_normalize(., timepoints = "day") :  
  No annotations are stored in this gimap_dataset, annotations are neededso we know what genes should be used as controls.Please run  
  gimap_annotate() function on your gimap_dataset and then retry this function.  
4.  
stop("No annotations are stored in this gimap_dataset, annotations are needed",  
  "so we know what genes should be used as controls.", "Please run gimap_annotate() function on your gimap_dataset and then",  
  " retry this function.")  
3.  
gimap_normalize(., timepoints = "day")  
2.  
calc_gi(.)  
1.  
gimap_dataset %>% gimap_filter() %>% gimap_annotate(cell_line = "HELA") %>%  
gimap_normalize(timepoints = "day") %>% calc_gi()
```

Site builder - consider adding a GitHub action to build the site on triggers (e.g. PRs to main, etc) using `pkgdown`. You currently build the site using `pkgdown` but store all of the HTML/JS intermediates on the main branch which will increase the payload size of this package.



fboehm on Feb 6

[@kalyanidhusia](#) please feel free to create your checklist



fboehm on Feb 14

[@kalyanidhusia](#) - please feel free to create your checklist and begin the review.



1

fboehm on Feb 14

[@cansavvy](#) - how are the revisions progressing? Please let me know if you have any questions on how to proceed.



|

16 remaining items

Load more

fboehm last month

[@cansavvy](#) - how are the revisions coming along? please feel free to reply in this thread.

Thanks!





fboehm last month

...

@Kevin-Mattheus-Moerman - What is the procedure for rejecting a JOSS submission? The submitting author has not replied to recent queries. Thanks.



cansavvy last month

...

@Kevin-Mattheus-Moerman - What is the procedure for rejecting a JOSS submission? The submitting author has not replied to recent queries. Thanks.

So sorry for the delay. I have switched institutions and roles and am no longer paid to work on this so it's taking me awhile to find free time



cansavvy last month

...

It sounds like my other coauthors are also unable to take on this paper so what's a final deadline you'd like me to have this addressed by and I can tell you whether or not I can meet that deadline?



fboehm last month

...

Thank you, @cansavvy. It would be great to address the reviewers' comments before April 30. If that is not possible, then we should consider whether to continue with the review process at this time.



cansavvy last month

...

Thank you, @cansavvy. It would be great to address the reviewers' comments before April 30. If that is not possible, then we should consider whether to continue with the review process at this time.

Okay thanks. Yes I think I can do before April 30th. Sorry for this delay.



1



cansavvy 3 weeks ago

...

I have raised a few issues in the the [target repository](#).

Thanks for your reviews and patience. It's much appreciated. I am no longer in this position where I created this software originally and am no longer paid to maintain it so finding free time is tricky.

Here's a summary of the changes I did in response to @tomsing1 's very thoughtful review.

- License: Swapped to BSD-style text in LICENSE; DESCRIPTION → file LICENSE; updated docs license pages/links.
- README: CRAN vs GitHub install; DocToc refreshed.
- Example data: Session cache under tempdir(); removed risky options() paths; safe delete_example_data() (known files, allowed dirs); tests added; fixed roxygen so gimap_annotate docs attach to the right function.
- run_qc: Dropped unused plots_dir; updated template, vignettes, tests, docs.
- DepMap / annotate / normalize: More robust metadata parsing; graceful fallback when DepMap fails (still get annotations); optional CN; gimap_normalize auto turns off normalize_by_unexpressed if TPM flags missing; vignette note.
- Tests: Adjusted annotate test for new normalize behavior; run_qc skips without

All tests pass successfully locally.



fboehm 2 weeks ago

...

[@tomsing1](#) [@btmonier](#) - please review the changes and, if possible, update the review checklists.

Thanks!



[fboehm](#) 2 weeks ago

[@btmonier](#) [@tomsing1](#) - I recognize that it has been some time since the start of this review. Would you please reply to let us know if you are still available to review this submission? Thanks again!



[btmonier](#) 2 weeks ago

[@fboehm](#) - yes, I should have time later this week or next week to take a look at the modifications.



[cansavvy](#) 2 weeks ago

Thanks to all of you including [@fboehm](#) and [@tomsing1](#) for your patience and understanding.



[fboehm](#) last week

[@tomsing1](#) - just wanted to check to see if you might have availability to review the latest package changes.

Thanks again!



[fboehm](#) 4 days ago

[@kalyanidhusia](#) - I just wanted to check in to see how the review is going. Are you available to check the latest updates from [@cansavvy](#) ?



[kalyanidhusia](#) 3 days ago · edited by kalyanidhusia

Edits ▾ ...

[@cansavvy](#)

I began a code-based review by installing the package and running the quick-start workflow using modules `get_example_data()`, and also reviewed the normalization `gimap_normalize` and DepMap annotation `depmap_metadata` functions and quick-start vignette: <https://fredhutch.github.io/gimap/articles/quick-start.html>

1. How I tested (for context)

- I cloned the repo locally
- Installed using both: `standard install (CRAN / GitHub)` and `local install from cloned repo`
- Ran the quick-start workflow end-to-end
- Also ran a small comparison test around normalization behavior

|| Let me know if it would be useful for me to share the small test scripts I used.

2. Small usability notes

- Input format clarity (`setup_data`)
I initially ran into input-shape errors when passing full example data directly. The workflow works well when following this pattern:
`select relevant columns > convert to matrix > derive pg_ids from same object`

Suggestion(i think It was pointed out before as well)

- ☐ Add one short note in docs that `counts` should be a matrix

☐ Maybe show `as.matrix()` explicitly in the example

code: https://github.com/FredHutch/gimap/blob/c8539aa0284c014c21c597ec237f7bb4eb377e0d/R/00-setup_data.R

3. Local install note (very minor)

- Installing from a cloned repo worked, but took a bit more troubleshooting than the standard install
- Suggestion:

☐ Add 1–2 lines in README for local `install` (e.g. `remotes::install_local()`)

4. Normalization behavior (small clarification request)

I ran a quick comparison:

default: `gimap_normalize(timepoints = "day")`

fallback: `normalize_by_unexpressed = FALSE`

For the example dataset, results were identical (GI scores and FDR unchanged)

suggestion

☐ Add a short note explaining when this parameter is expected to matter, or

☐ Mention if example dataset is intentionally insensitive

code: <https://github.com/FredHutch/gimap/blob/c8539aa0284c014c21c597ec237f7bb4eb377e0d/R/04-normalize.R>

5. Annotation / DepMap

The fallback behavior works smoothly. Just a small suggestion to clarify it for users:

☐ Add a short explanation of fallback behavior in vignette

code: <https://github.com/FredHutch/gimap/blob/c8539aa0284c014c21c597ec237f7bb4eb377e0d/R/03-annotate.R>

Happy to help!



btmonier 7 minutes ago

[@fboehm](#) - I have updated my review checklist. The current blocker is the references. [@cansavvy](#) - if you could update these that would be great. I have added these as [itemized issues](#) labelled as "JOSS review".



cansavvy 1 minute ago

Will address these in the next couple days. Thanks.



Add a comment

Write

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Comment

Assignees

-  fboehm
-  tomsing1

Labels

- HTML
- R
- TeX
- Track: 2 (BCM)
- pre-2026-submission
- review

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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